

AI-Assisted Data Analysis Using Agentic AI

Coffee Quality Dataset Analysis

I chose Claude as my AI platform to analyze a public dataset from Kaggle that provides information on coffee quality around the world and generate insights, visualizations, and a short analytical report.

1. Dataset Description

1.1 Source

The dataset was sourced from Kaggle (df_arabica_clean.csv) and contains arabica coffee quality reviews collected by the Coffee Quality Institute (CQI).

1.2 Size

The dataset contains 207 rows and 41 columns, each row representing a unique coffee sample.

1.3 Main Features/Variables

The dataset covers five main categories of variables:

- Origin & Identity: Country of Origin, Farm Name, Region, Producer, Variety, Processing Method
- Logistics: Harvest Year, Altitude, Number of Bags, Bag Weight
- Taste Scores (scale ~6.5-10): Aroma, Flavor, Aftertaste, Acidity, Body, Balance, Overall, Total Cup Points
- Defects: Category One Defects, Category Two Defects, Quakers
- Certification: Certification Body, Certification Address, Certification Contact

2. AI-Assisted Analysis

I used Claude to guide me through each step of the analysis from exploring the dataset structure to generating visualizations. Claude suggested the code and approaches, and I verified the results against the actual data.

2.1 Explore the Dataset

I started by exploring the dataset to understand its structure and the distribution of categorical variables such as country of origin and processing method. I also checked whether any owner submitted more than one variety of coffee, finding that Coffee Quality Union leads with 14 distinct varieties and Taiwan Coffee Laboratory with 12.

2.2 Detect Missing Values

I asked Claude to help me detect missing values across the dataset. I found that most columns are complete, but a few have notable gaps particularly ICO Number, which is missing for the majority of entries.

What is an ICO Number? *The ICO (International Coffee Organization) Number is a tracking code assigned to coffee lots for international trade and export purposes. Its absence in most rows suggests many samples were submitted outside the formal ICO export pipeline, or the field was simply not recorded.*

2.3 Generate Statistics

I used Claude to generate descriptive statistics across several dimensions taste scores, country averages, and defect counts.

2.3.1 Taste Score Statistics

The taste scores are fairly tightly distributed, with Total Cup Points ranging from 78 to 89.33 and a mean of around 83.7.

2.3.2 Average Score by Country and Continent

I calculated the average Total Cup Points per country to see which origins produce the highest-rated coffee.

2.3.3 Defect Frequency

I explored how often defects appear across samples.

What are Category One Defects? *Primary (full) defects severe issues like black beans, sour/fermented beans, fungus-damaged beans, or foreign matter. Each instance counts as one full defect and has a strong negative impact on quality.*

What are Category Two Defects? *Secondary (partial) defects less severe issues like immature beans, broken or chipped beans, floaters, parchment, or slight insect damage. Multiple affected beans may be needed to count as a single defect.*

These are not dataset-specific terms they are a globally standardized grading system defined by the Specialty Coffee Association (SCA). To qualify as specialty grade, a coffee must have zero Category One defects and fewer than 5 Category Two defects in a 350g sample.

Category One defects are rare 93% of samples have none at all. Category Two defects are more common and vary widely, with some samples having as many as 16.

2.4 Find Correlations and Trends

I asked Claude to help me identify which variables are most strongly related to the overall quality score.

2.4.1 Taste Score Correlations

The results showed that Overall, Flavor, and Aftertaste have the strongest positive relationships with Total Cup Points, with Overall scoring the highest correlation at 0.947, followed by Flavor at 0.939 and Aftertaste at 0.935.

2.4.2 Quakers and Their Effect on Score

What are Quakers? *Quakers are underdeveloped coffee beans that fail to roast properly, staying pale or yellowish while other beans darken. They produce flat, peanut-like, or bland flavors and are considered a defect in specialty coffee grading.*

I investigated whether the number of quakers affects the Total Cup Points. The correlation came out at -0.32, a weak negative relationship. Samples with 0 quakers averaged 83.88 points, while samples with 12 quakers dropped to 78.08 showing that higher quaker counts tend to lower the score, though the effect is moderate.

2.5 Visualizations

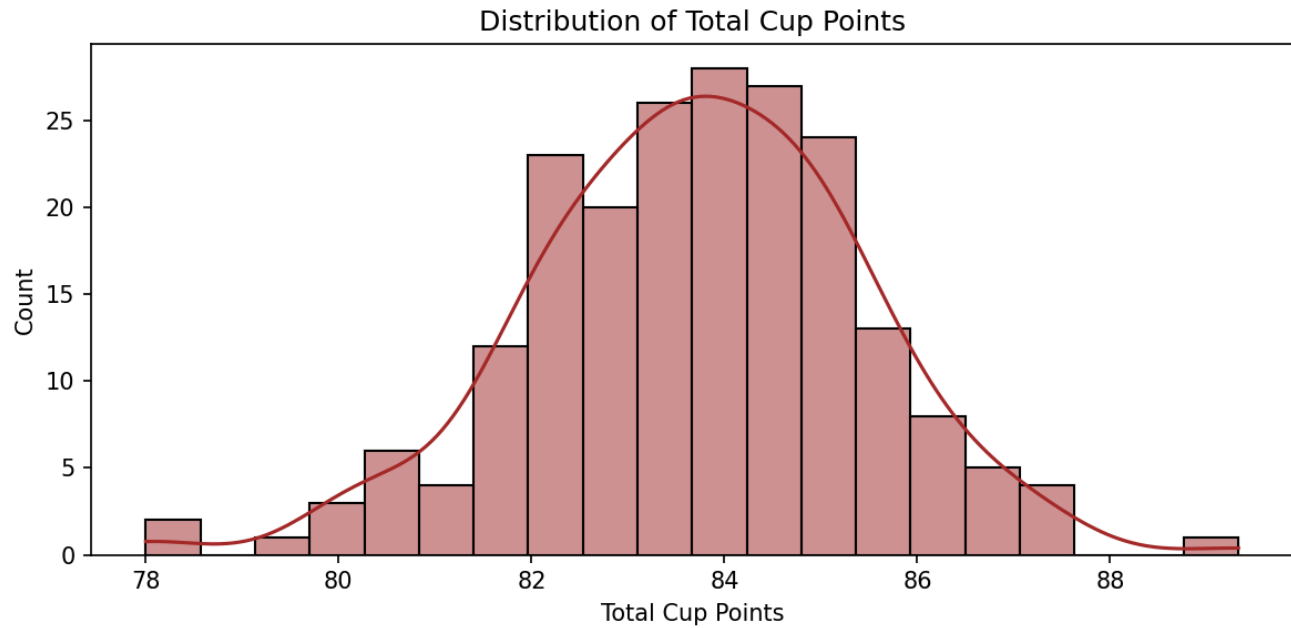


Figure 1: Distribution of Total Cup Points

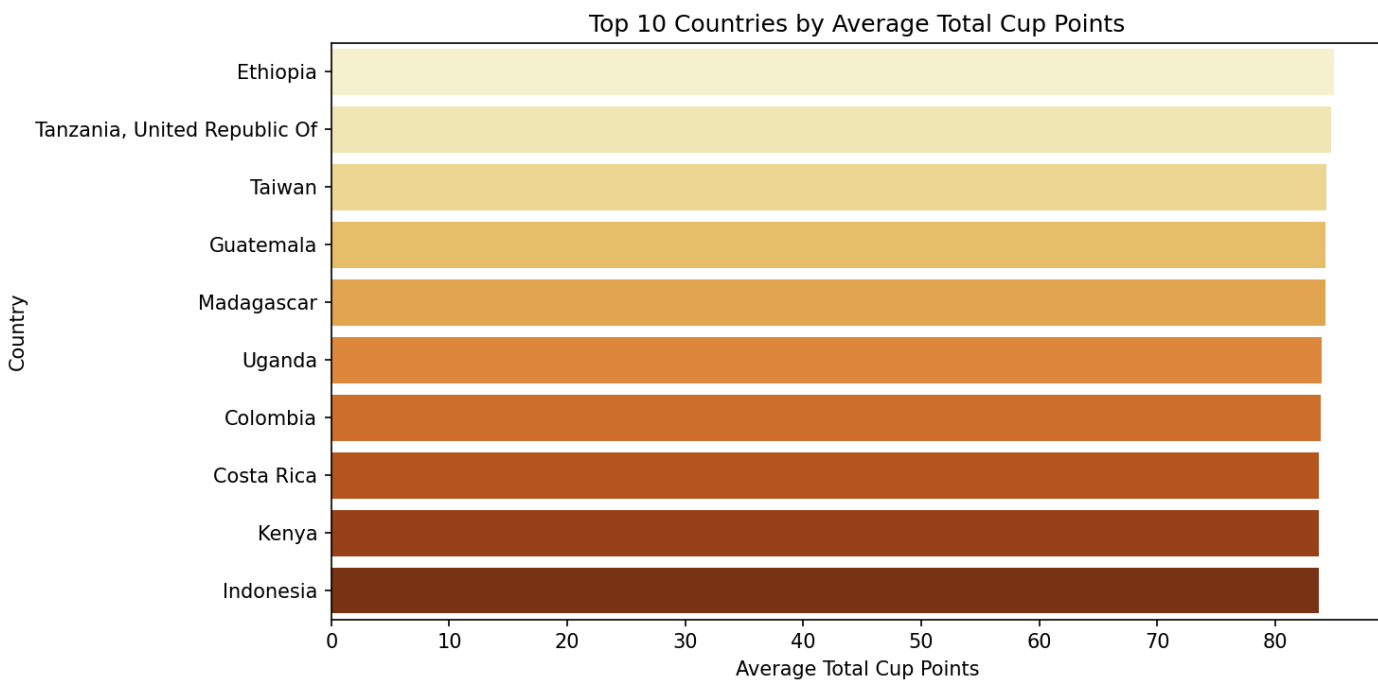


Figure 2: Top 10 Countries by Average Total Cup Points

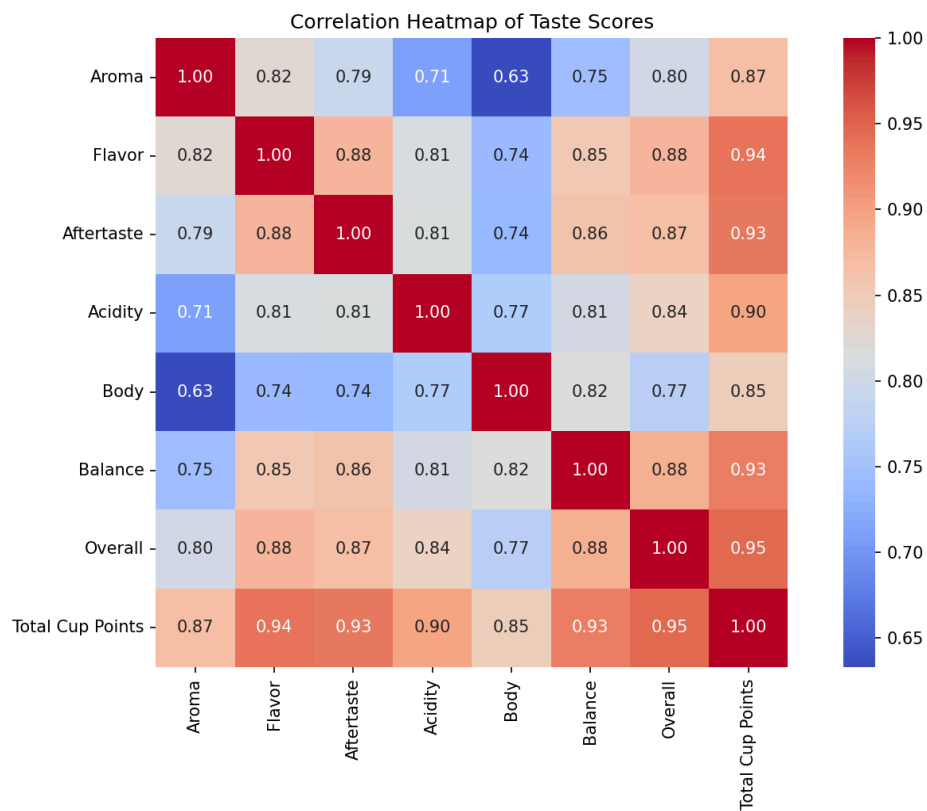


Figure 3: Correlation Heatmap of Taste Scores

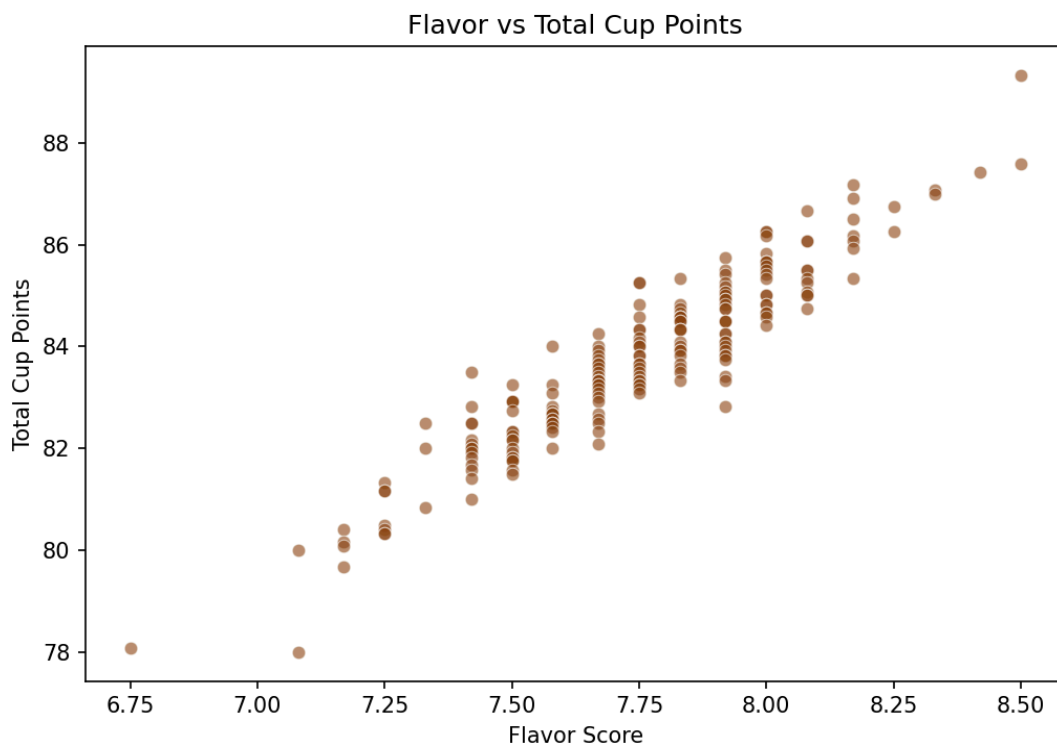


Figure 4: Flavor vs Total Cup Points

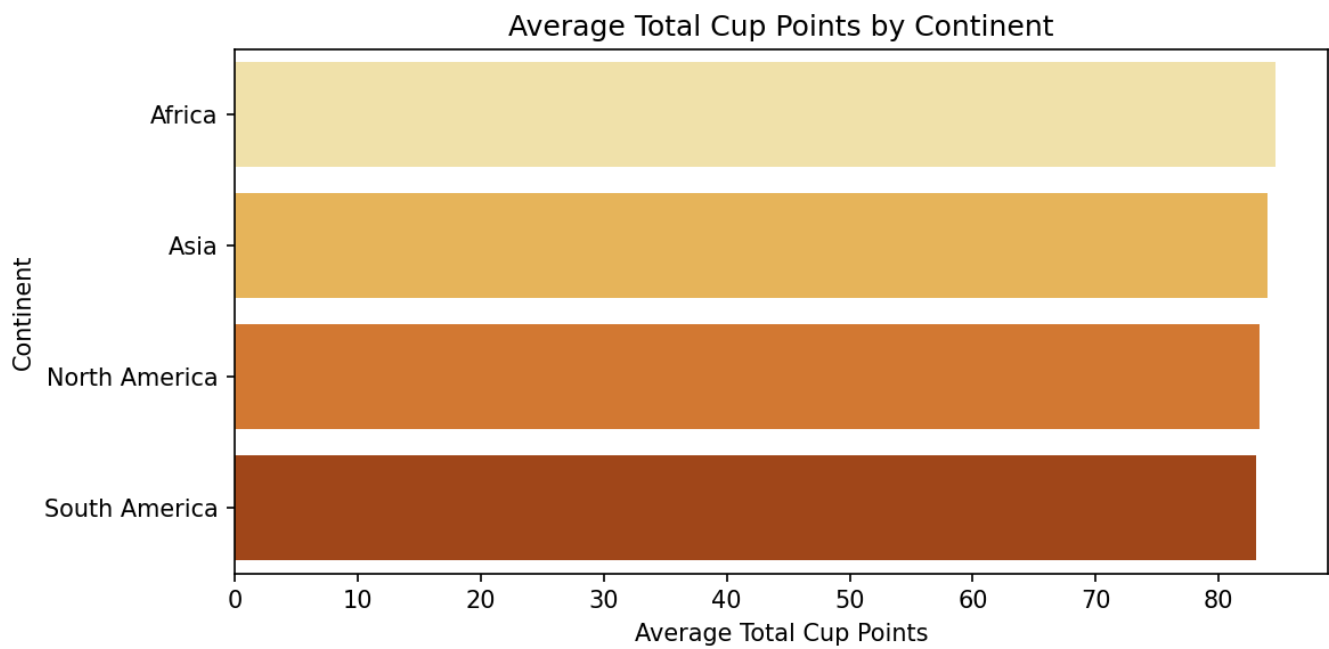


Figure 5: Average Total Cup Points by Continent

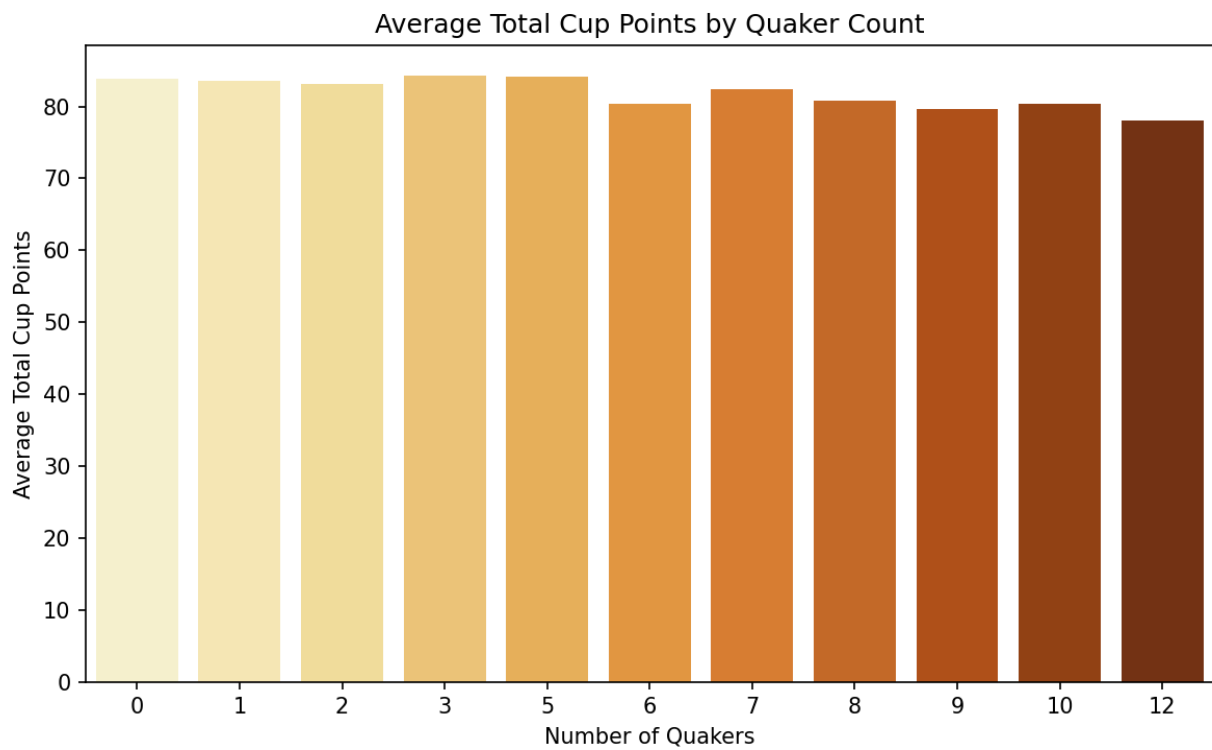


Figure 6: Average Total Cup Points by Quaker Count

3. Insights

After completing the analysis with Claude's assistance, I identified the following key insights.

Insight 1: Overall Score is the Strongest Predictor of Quality

Among all taste attributes, Overall has the highest correlation with Total Cup Points at 0.947, followed closely by Flavor (0.939) and Aftertaste (0.935). This means that the grader's holistic impression of the coffee drives the final score more than any single attribute. Flavor remains the most influential individual taste dimension.

Insight 2: Scores Are Tightly Clustered High-Scorers Stand Out

The majority of samples score between 82 and 86, with a mean of 83.7. The distribution is slightly left-skewed, meaning a few exceptional samples pull the average up. Any sample scoring above 87 is a clear outlier and represents top-tier specialty coffee.

Insight 3: ICO Number Has Too Many Missing Values to Be Useful

The ICO Number column is missing for 63.77% of entries (132 out of 207). This makes it unreliable for any analysis or filtering and should be excluded from any model or further analysis.

Insight 4: Category One Defects Are Rare Category Two Are Not

93% of samples have zero Category One defects, suggesting that major defects are screened out before grading. However, Category Two defects appear in the majority of samples and range from 0 to 16. This makes Category Two defects a more meaningful quality differentiator.

Insight 5: Quakers Negatively Affect the Score

There is a weak negative correlation (-0.32) between quaker count and Total Cup Points. Samples with no quakers averaged 83.88 points while samples with 12 quakers dropped to 78.08. Although the relationship is not strong, it confirms that underdeveloped beans consistently drag down the overall quality score.

Insight 6: A Few Large Organizations Dominate the Dataset

Coffee Quality Union submitted 14 different varieties and Taiwan Coffee Laboratory submitted 12, while most other owners submitted only one or two. This suggests these are large-scale certification organizations rather than individual producers. It also made me realize that much of the competition in the dataset could be happening within or between just two or three major companies, creating a kind of internal hierarchy where the same organizations are both submitting and dominating the top-scoring entries.

Insight 7: Rank Of Continent & Country

I mapped each country to its continent and compared average Total Cup Points. Africa comes out on top with an average of 84.63, driven by strong performances from Ethiopia (84.96) and Tanzania (84.74). Asia follows at 84.00, largely led by Taiwan (84.35). North America and South America trail behind at 83.33 and 83.09 respectively. At the country level, Ethiopia ranks first with an average of 84.96, making it the highest-rated origin in the dataset. El Salvador sits at the bottom with 81.53. It is worth noting that some countries like Madagascar and Myanmar only have one sample each, so their averages are not statistically reliable.

4. Reflection

4.1 Which AI Tool I Used

Throughout this project, I used Claude by Anthropic as my AI assistant. Before writing a single line of code, Claude reviewed the dataset with me to confirm it contained enough varied and valuable data to support a meaningful analysis. Once I confirmed the dataset was worth working with, Claude guided me through each stage, not by completing tasks for me, but by helping me understand what was needed and how to approach it.

4.2 Advantages

- **Dataset validation upfront:** Claude reviewed the CSV file at the start to identify which columns were most valuable and flag any potential issues before I began coding.
- **Code collaboration:** I wrote the code myself, and Claude helped me extend and combine it where needed. For example, building the country-to-continent mapping that allowed me to extract continent-level insights from country-level data.
- **Domain knowledge on demand:** Claude explained terms like quakers, SCA grading standards, and ICO numbers as we worked without me having to look them up.
- **Grammatical and structural feedback:** Claude reviewed my markdown cells and report writing, helping me fix grammatical mistakes and keep the language consistent.
- **Project structure:** Claude helped me maintain a clean and logical folder structure across notebooks, data, visuals, and the report from the start.
- **Iterative collaboration:** I could ask follow-up questions, request changes, and correct mistakes mid-session and Claude would adjust without losing context.

4.3 Limitations

- **Cannot execute code:** Claude is unable to run cells. Every cell had to be executed by me in Jupyter, and I was responsible for verifying that outputs matched what was expected.
- **No visual feedback:** Claude cannot see generated charts after they are saved so I had to open and review every visualization in the visuals folder myself to confirm it was correct.
- **File corruption risk:** On several occasions, editing the notebook file directly through the terminal caused JSON corruption, requiring the file to be fully rebuilt from scratch.
- **Configuration limitations:** Claude could not resolve environment-level issues directly. For example, when Jupyter was truncating output rows by default, Claude could only advise me on what settings to change and I had to apply the fix myself.
- **Requires active involvement:** The collaboration only worked because I stayed engaged at every step. Claude assisted, but every decision, verification, and final judgment was mine.

4.4 How I Verified AI-Generated Results

- Every code cell was executed in Jupyter and I compared the output to what Claude had described before accepting it.
- For key statistics including the mean Total Cup Points, the top-ranked country, and all correlation values, I re-ran the relevant cells independently to confirm accuracy.
- Every chart saved to the visuals folder was opened and reviewed visually to confirm the data, labels, and titles were accurate and matched the analysis.